

Computer Graphics

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Assignment 3

Implementing Ray Tracer

- Objective
 - The goal of this assignment is to write a simple ray tracer which can render a simple scene consisting of a plane and three spheres, and shade them using the Phong model.



Questions

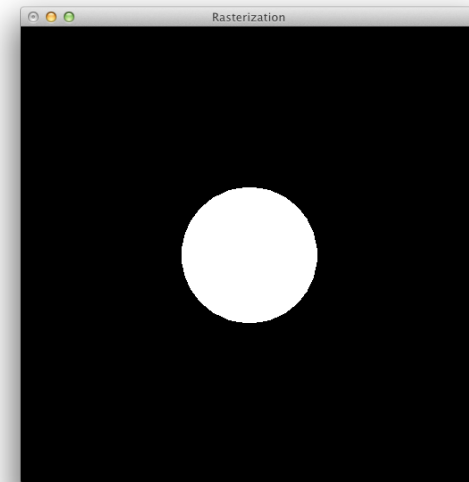
Q1: Ray Intersection

- The scene consists of the following four objects:
 - Plane P , with equation $y = -2$.
 - Sphere S1, with center at $(-4, 0, -7)$ and radius 1.
 - Sphere S2, with center at $(0, 0, -7)$ and radius 2.
 - Sphere S3, with center at $(4, 0, -7)$ and radius 1.
- Camera
 - Assume a perspective camera
 - eye point at $e = (0, 0, 0)$
 - orientation given by $u = (1, 0, 0)$, $v = (0, 1, 0)$ and $w = (0, 0, 1)$.
 - (Note that the camera is looking along the direction $-w$.)
 - Assume that the viewing region on the image plane is defined by $l = -0.1$, $r = 0.1$, $b = -0.1$, $t = 0.1$, and $d = 0.1$.
 - Also assume that the image resolution is 1024×1024 (i.e., $n_x = n_y = 1024$).



Q1: Ray Intersection

- Ray Tracer
 - Write a ray tracer which generates eye rays through the center of each pixel
 - Calculates the intersection between each ray and each object in the scene
 - Also calculating the closest intersection (if any) for each ray.
 - For each pixel, if the corresponding ray intersects an object, set the pixel's color to white; otherwise, set the pixel's color to black.
 - See the Figure for a reference image.



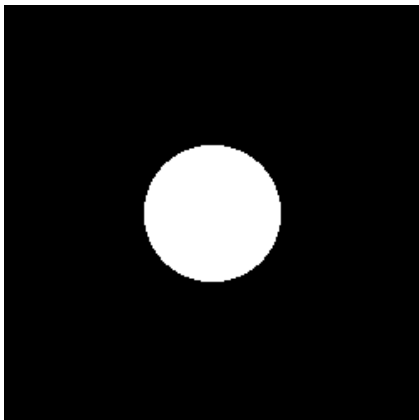
The output must be different from the image shown here. 7



Q1: Ray Intersection

- Implement the Following Classes:

- Ray
- Camera
- Surface
- Plane (inherits from Surface)
- Sphere (inherits from Surface)
- Scene
- Any additional classes as needed



```
Surface s = new Sphere(0.0, 0.0, 0.0), 1.0);  
for 0 <= iy < ny  
    for 0 <= ix < nx {  
        ray = camera.getRay(ix, iy);  
        c = scene.trace(ray, 0, +inf);  
        image.set(ix, iy, c);  
    }
```

...

```
Scene.trace(ray, tMin, tMax) {  
    surface, t = surfs.intersect(ray, tMin, tMax);  
    if (surface != null) return white;  
    else return black;  
}
```


Q2: Phong Shading

- Use the scene and camera settings from Q1.
- Use the Phong model to shade the scene, with the following material parameters for each object:
 - Plane P : $k_a = (0.2, 0.2, 0.2)$, $k_d = (1, 1, 1)$, $k_s = (0, 0, 0)$, with specular power 0.
 - Sphere S1: $k_a = (0.2, 0, 0)$, $k_d = (1, 0, 0)$, $k_s = (0, 0, 0)$, with specular power 0.
 - Sphere S2: $k_a = (0, 0.2, 0)$, $k_d = (0, 0.5, 0)$, $k_s = (0.5, 0.5, 0.5)$, with specular power 32.
 - Sphere S3: $k_a = (0, 0, 0.2)$, $k_d = (0, 0, 1)$, $k_s = (0, 0, 0)$, with specular power 0.
- Note that the k_a , k_d , and k_s values are specified as (red, green, blue) values.



Q2: Phong Shading

- Assume a single **point light source** at $(-4, 4, -3)$, emitting white light with unit intensity and no falloff.
 - Light Intensity: $I = ?$
 - Light Direction: $\mathbf{l} = ?$
- Assume a white ambient light with unit intensity.
 - Ambient Light: $I_a = ?$
- Modify your ray tracer to store at each pixel the appropriate illumination values computed using the Phong shading model.
- Implement shadows from the point light using shadow rays.



Q3: Implement a New Feature

- Implement and demonstrate a custom feature
 - e.g., Drawing differently shaped objects (other than planes and spheres)
 - e.g., Adding more light sources
 - e.g., Moving objects over time



Instructions

Submission Guidelines

- Important Notes

- Your code must include a single project for each question
 - In this assignment, your VS2022 solution must include three projects
- You are not allowed to use OpenGL to perform hardware rasterization. You must implement your own ray tracing and shading algorithms.
 - The provided sample code uses a small amount of OpenGL and GLUT to display your image on-screen.
- With your code, please submit output screenshots for each question and explain them during your presentation

- Submission Requirements

- Presentation video: Under 5 minutes and 30MB maximum (MP4 format)
- Presentation slides: PDF format
- Source code + output screenshots: VS 2022 project and image files (ZIP format)



Summary

- Assignment 3
- Questions
- Instructions



End of Class

Thank you

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